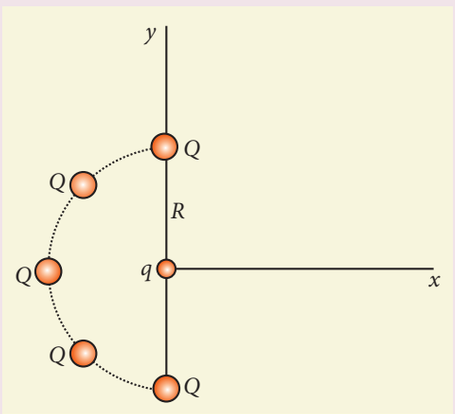
and your friend lost 1% of this number of electrons. Calculate the electrostatic force between you and your friend separated at a distance of 1m. Compare this with your weight. Assume mass of each person is 60 kg and use point charge approximation.

Ans: $F_e = 23 \times 10^{23}$ N, W = 588 N,

$$\frac{F_e}{W} = 3.9 \times 10^{21}$$

3. Five identical charges Q are placed equidistant on a semicircle as shown in the figure. Another point charge q is kept at the centre of the circle of radius R. Calculate the electrostatic force experienced by the charge q.



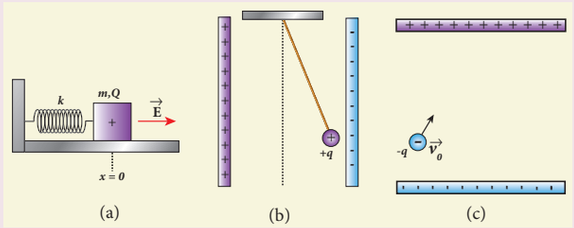
Ans: $\vec{F} = \frac{1}{4\pi\epsilon_0}\frac{qQ}{R^2}\left(1+\sqrt{2}\right)\hat{i}$ N

4. Suppose a charge $+q$ on Earth's surface and another $+q$ charge is placed on the surface of the Moon. (a) Calculate the value of q required to balance the gravitational attraction between Earth and Moon (b) Suppose the distance between the Moon and Earth is halved, would the charge q change?

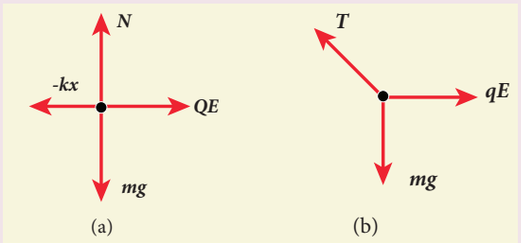
(Take $m_E = 5.9 \times 10^{24}$ kg, $m_M = 7.9 \times 10^{22}$ kg)

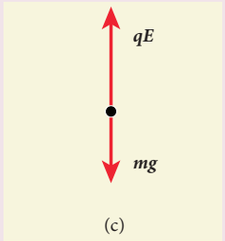
Ans: (a) $q \approx +5.87 \times 10^{13}$ C, (b) no change

5. Draw the free body diagram for the following charges as shown in the figure (a), (b) and (c).

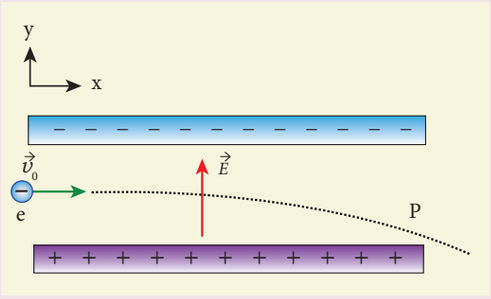


Ans:





6. Consider an electron travelling with a speed v_0 and entering into a uniform electric field $\vec{E}$ which is perpendicular to $\vec{v}_0$ as shown in the Figure. Ignoring gravity, obtain the electron's acceleration, velocity and position as functions of time.



Ans :

$$\vec{a} = -\frac{eE}{m}\hat{j}, \vec{v} = v_0\hat{i} - \frac{eE}{m}t\hat{j}, \vec{r} = v_0 t\hat{i} - \frac{1}{2}\frac{eE}{m}t^2\hat{j}$$

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